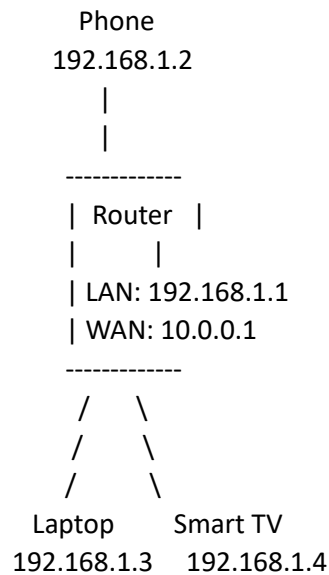


N+ - Network Fundamentals and building network

IP Routing & Routing Protocols

[Q-1] Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



LAN Network: 192.168.1.0/24

All devices are connected to the same local area network (LAN).

[Q-2] Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.
Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Example

- Laptop IP: 192.168.1.3
- Friend's PC IP: 192.168.2.5
- Router LAN Interface: 192.168.1.1
- Router WAN Interface: 192.168.2.1

Step-by-Step Process

1. The laptop wants to send data to 192.168.2.5.

2. It checks the destination IP address and determines that it belongs to a different subnet.
3. The laptop sends the packet to its default gateway (192.168.1.1).
4. The router receives the packet.
5. The router checks its routing table.
6. It finds that network 192.168.2.0/24 is reachable through its WAN interface.
7. The router forwards the packet to the destination network.
8. Friend's computer (192.168.2.5) receives the packet.

Final Path

```

Laptop (192.168.1.3)
      v
Router (192.168.1.1)
      v
Friend PC (192.168.2.5)
  
```

[Q-3] Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Destination Network	Subnet Mask	Next Hop	Interface
10.1.0.0	255.255.255.0	Direct	Local
10.2.0.0 (Delhi)	255.255.255.0	10.1.0.2	Delhi
10.3.0.0 (Mumbai)	255.255.255.0	10.1.0.3	Mumbai

Meaning

- Delhi, Mumbai, and Bangalore represent different delivery hubs.
- Similar to how Zomato chooses the best delivery route, a router selects the best path for forwarding packets.
- The router uses the routing table to decide where data should be sent next.

[Q-4] Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing

Advantages

- Easy to configure in small networks.
- No routing update traffic.
- More secure because routes are manually controlled.

Disadvantages

- Requires manual updates.
- Difficult to manage in large networks.
- Route failures require administrator intervention.

Dynamic Routing (RIP, OSPF)

Advantages

- Routes are updated automatically.
- Suitable for large networks.
- Automatically adapts to network changes and failures.

Disadvantages

- More complex configuration.
- Consumes CPU and memory resources.
- Generates routing update traffic.

[Q-5] Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
```

Explanation

Command	Purpose
router ospf 1	Starts OSPF process ID 1
network 192.168.1.0 0.0.0.255 area 0	Advertises network 192.168.1.0/24 in OSPF
0.0.0.255	Wildcard mask for a /24 network

Command

Purpose

area 0

Backbone area of OSPF

network 192.168.2.0 0.0.0.255 area 0 Advertises network 192.168.2.0/24 in OSPF